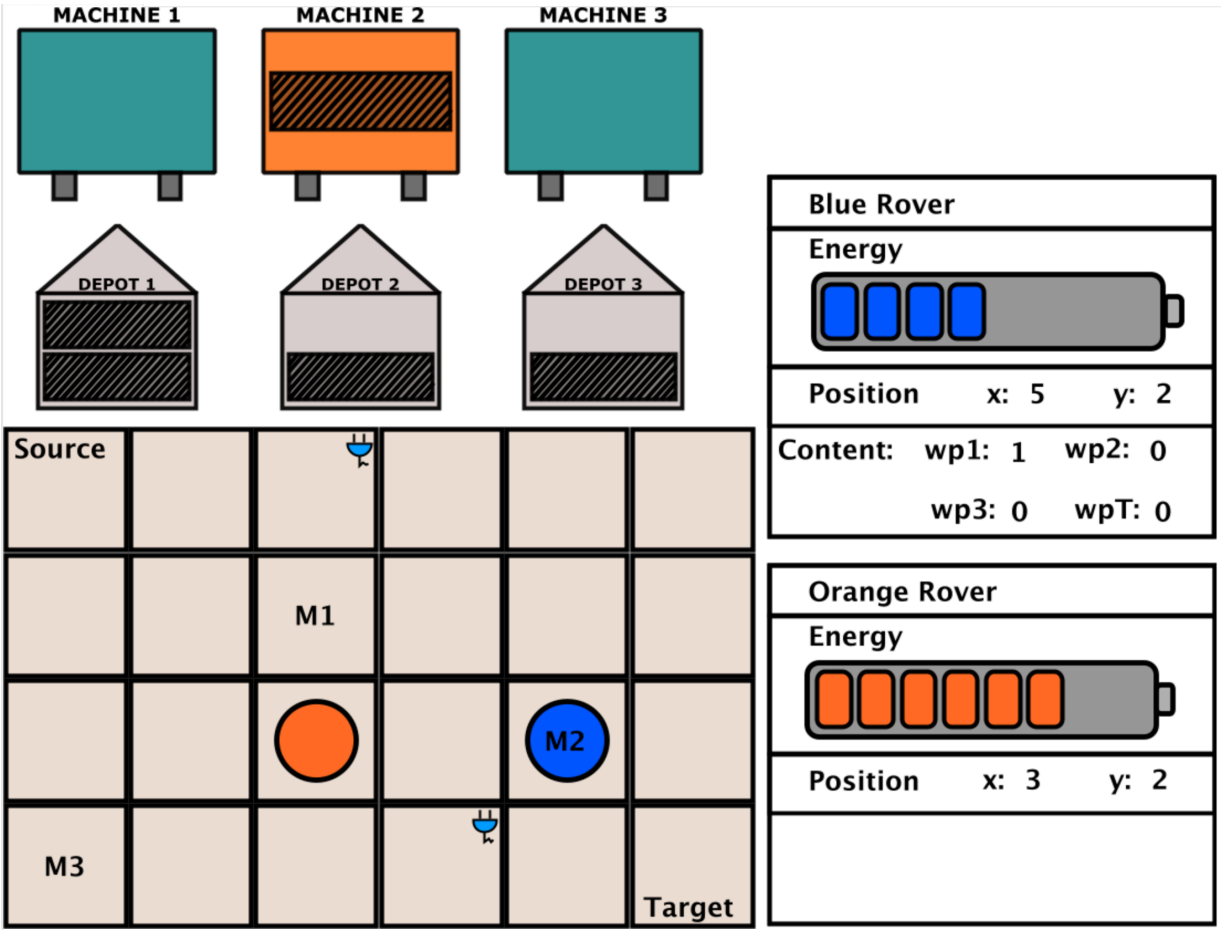
Data limite: venerdì, 6 febbraio 2026, 23:59



Description of the plant:

The figure above provides a graphical representation of a plant in which two rovers (blue and orange circles) move in a factory area represented as a (6×4) grid, where $x = 1, y = 1$ is the bottom-left tile, and $x = 6, y = 4$ is the top-right tile.

The aim of the factory is to work workpieces w. The blue rover is in charge of moving workpieces between a few points of interest according to the following workflow.

1. The blue robot picks a workpiece w from an infinite source of workpieces located at tile $x = 1, y = 4$.
2. The blue robot takes the workpiece w from the source to machine M_1.
3. Machine M_1 works w and either deposits the workpiece in depot D_1 (if it finishes successfully) or discards it (if the machine breaks down). Machine M_1 and depot D_1 are located at tile $x = 3, y = 3$.
4. The blue robot picks a workpiece w from depot D_1 and moves it to machine M_2.
5. Machine M_2 works w and either deposits the workpiece in depot D_2 (if it finishes successfully) or discards it (if the machine breaks down). Machine M_2 and depot D_2 are located at tile $x = 5, y = 2$.
6. The blue robot picks a workpiece w from depot D_2 and moves it to machine M_3.
7. Machine M_3 works w and either deposits the workpiece in depot D_3 (if it finishes successfully) or discards it (if the machine breaks down). Machine M_3 and depot D_3 are located at tile $x = 1, y = 1$.
8. The blue robot picks a workpiece w from depot D_3 and moves it to an infinite capacity target located at tile $x = 6, y = 1$.

All depots can store up to two workpieces.

The orange rover is in charge of repairing the machines in case they break down. If some machine M_i breaks down, the orange rover needs to get to M_i (=reach the specific tile) in order to repair it.

Moreover, the blue and orange rover:

- are equipped with a battery with a capacity of 8 units of energy;
- starts with a fully charged battery;
- can move up, down, left, or right provided they stay within the grid boundaries and the battery is not empty;
- consume a unit of energy anytime they make a move.

The blue rover can also carry workpieces by means of some internal storage. Specifically,

- it can carry up to 2 workpieces picked from the source;
- it can carry up to 2 workpieces picked from D_1;
- it can carry up to 2 workpieces picked from D_2;
- it can carry up to 2 workpieces picked from D_3.

Finally, two charging stations are located at tiles $x = 4, y = 1$ and $x = 3, y = 4$. If a rover sits on any of these tiles, then it can get a full charge.

Tasks:

1. Design a suitable SVG graphical interface for the plant.
2. Design the plant modules and describe your choice for the marking.
3. Enforce the following control policies. You are free to decide if to enforce them at plant level or at requirement level, provided that at least 3 of them are formalized as requirements.
   - R_1: no rover runs out of battery on tiles that are not charging stations;
   - R_2: the sums of the units of energy of the two batteries is always ≥ 1;
   - R_3: the maximum number of workpieces that the blue rover can carry is 4;
   - R_4: each rover can charge provided its battery is not already full;
   - R_5: rovers do not collide;
   - R_6: if D_3 is full, then D_1 can store at most one workpiece.
4. Write a tooldef script to synthesize a supervisor enforcing all requirements;
5. Write a PDF report justifying all of your design choices.

Organize the files as follows:

- **a root folder** named with the surnames of all group members (e.g., **bresolin-zavatteri**);
- **a subfolder escet** with all CIF/tooldef/SVG implementation files and directories (e.g., **bresolin-zavatteri/escet**);
- a report in PDF (max 10 pages, format A4) inside the root folder (e.g., b**resolin-zavatteri/report.pdf**).

Submission:

Compress the root folder and submit the result (e.g., **bresolin-zavatteri.zip**).

Stato consegna

Il compito prevede la consegna di gruppo ma tu non sei membro di nessun gruppo e non puoi effettuare consegne. Per favore contatta il docente affinché ti inserisca in un gruppo. ×

Gruppo	Non appartieni a nessun gruppo
Stato consegna	Non sono presenti consegne da valutare
Stato valutazione	Non valutata
Tempo rimasto	Consegna in ritardo da: 85 giorni 11 ore